Md Kamrul Islam

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Google Scholar | ORCID | Portfolio | GitHub | LinkedIn

Research Interests

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multimodal AI, and Information Retrieval.

Research Experience

AI Research Engineer Intern [Project] *May 2025 – Oct 2025*
LISSI, Université Paris-Est Créteil | in collaboration with CEA–LIST *Paris, France*

- Developed a hybrid framework combining LLM-based reasoning with rule-based validation for automated security annotation generation in Business Process Model and Notation (BPMN).
- Constructed a benchmark dataset and evaluated multiple LLMs using different prompt settings and RAG.
- Evaluated the framework against domain-expert annotations, achieving 0.58 precision and 0.52 recall while reducing annotation time by approximately 95%.

Methods & Tools: *LLMs, RAG, Python, LangChain, Flask, React.js, CI/CD*

Graduate Research Assistant [Project] *Oct 2024 – Mar 2025*
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec *Paris, France*

- Developed a deep-clustering framework combining Convolutional Neural Networks (CNNs) and Group-Equivariant CNNs (G-CNNs) for a large-scale chest X-ray dataset.
- Investigated geometric inductive biases to improve feature representation and reduce reliance on data augmentation.

Methods & Tools: *PyTorch, G-CNNs, OpenCV, CUDA, HPC*

Academic & Professional Experience

Lecturer [Faculty Profile] *April 2026 – Present*
Department of Computer Science, American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- Teach undergraduate courses in Data Processing, Data Visualization, and Programming for Data Science.
- Develop course materials, laboratory exercises, and project-based learning activities for undergraduate courses.
- Supervise undergraduate research projects in problem formulation, experimentation, and scientific writing.

Big Data Engineer Intern *Dec 2020 – May 2021*
Chengdu Suncape Co., Ltd. *Chengdu, China*

- Developed an end-to-end data pipeline for large-scale data ingestion, transformation, and analytics in distributed environments.
- Enhanced data preprocessing and feature engineering, improving predictive model accuracy by 20%.
- Collaborated in an Agile Scrum environment to implement modular software design and CI/CD practices.

Methods & Tools: *Apache Spark, Scikit-learn, SQL, Git, Jira, CI/CD*

Publications

1. **M. K. Islam**, T. Henry, M. Salnitri, J. Köpke, and S. Souihi, "A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models," in *Proceedings of the 24th International Conference on Business Process Management (BPM 2026), Business Process Management Forum*, Toronto, ON, Canada, 2026, pp. 168–184. *Lecture Notes in Business Information Processing*, vol. 599, Springer, Cham. [DOI]
2. C. K. Sah, L. Xiaoli, M. M. Islam, and **M. K. Islam**, "*Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education*," *36th International Conference on Software Engineering Education and Training (CSEE&T)*, Würzburg, Germany, 2024, pp. 1–5. [DOI]

Education

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA) *Sept 2023 – Oct 2025*
CentraleSupélec | Universitat Politècnica de Catalunya | Université libre de Bruxelles *France | Spain | Belgium*

- **Grade:** 14.5/20 (Bien/Good)
- **Thesis:** *A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models*

Bachelor of Engineering in Software Engineering *Mar 2018 – Dec 2021*
Sichuan University *Chengdu, China*

- **CGPA:** 3.64/4.00 (Weighted Average: 87%)
- **Thesis:** *Brain Tumor Detection and Classification Using Convolutional Neural Networks*

Technical Skills

Programming: Python, C/C++, SQL, JavaScript
AI/ML: PyTorch, Scikit-learn, OpenCV, Hugging Face Transformers, LLM APIs, LangChain
Big Data & Cloud: Apache Spark, Microsoft Azure, Microsoft Fabric, Apache Airflow
Databases: PostgreSQL, Neo4j, GraphDB, SPARQL
Computing: Linux, Docker, CUDA, HPC, Git

Projects

DigiScan360 [Project] *Feb 2024 – June 2024*
Developed a competitive intelligence platform and pitched it as a startup prototype.
Tech Stack: *PySpark, LLMs, Microsoft Fabric, Microsoft Azure, Power BI, GraphDB, SPARQL*

Academic Service & Activities

Peer Reviewer – 4th International Conference on Computing Advancements, 2026 [Certificate] *Aug 2026*
ACM SIGAPP In-Cooperation | American International University-Bangladesh *Dhaka, Bangladesh*

Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) [Poster] *July 2024*
University of Padova *Padova, Italy*

Honors & Scholarships

Erasmus Mundus Scholarship *2023 – 2025*
European Commission – Fully funded merit-based scholarship for the BDMA Joint Master's Programme

Belt and Road Initiative Scholarship *2018 – 2022*
Chinese Government – Scholarship worth CNY 80,000 over four years for undergraduate study

Languages

English: Proficient | **Chinese:** Upper-intermediate | **French:** Elementary | **Bengali:** Native

References

References available upon request.